

Red Hat Lightspeed 1-latest

Assessing and reporting malware signatures on RHEL systems with FedRAMP

Know when systems in your Red Hat Enterprise Linux infrastructure are exposed to malware risks

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Know when systems in your Red Hat Enterprise Linux infrastructure are exposed to malware risks

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Abstract

Use the Red Hat Lightspeed malware detection service in FedRAMP® environments with hundreds of IBM X-Force threat intelligence signatures to know when a system in your infrastructure is the victim of a malware attack.

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CHAPTER 1. RED HAT LIGHTSPEED MALWARE DETECTION SERVICE

The Red Hat Lightspeed malware detection service scans Red Hat Enterprise Linux systems for malware using YARA software with IBM X-Force threat intelligence signatures in partnership with Red Hat.

1.1. MALWARE DETECTION SERVICE CAPABILITIES

With the malware detection service, you can review signatures, analyze scan results, and identify malware based on your administrator or viewer permissions in the Red Hat Hybrid Cloud Console.

- Review the list of IBM X-Force signatures that the service uses to scan your RHEL systems.
- Review the overall results for all RHEL systems with malware detection enabled in the insights-client.
- Review the scan results for individual RHEL systems.
- Identify RHEL systems showing evidence of the presence of malware.

Use malware detection service findings to assess threats and coordinate incident response for affected RHEL systems.



IMPORTANT

The malware detection service does not provide recommendations to resolve or remediate malware incidents.

Additional resources

- Interactive demo of the malware service
- About YARA

1.2. MALWARE SIGNATURE SOURCES

YARA signature detection is the cornerstone of the Red Hat Lightspeed malware detection service. A YARA signature describes a malware type as a pattern: a set of strings and a boolean expression that form a rule. When a scan on a Red Hat Enterprise Linux system matches that rule, YARA records a hit on that system.

The malware detection service includes predefined signatures from the IBM X-Force Threat Intelligence team and evaluates your systems against hundreds of those default rules to help detect malware on Red Hat Enterprise Linux systems.

CHAPTER 2. MANAGE USER PERMISSIONS FOR RED HAT LIGHTSPEED SERVICES

Manage user permissions to control access to Red Hat Lightspeed applications. Use the User Access feature to apply role-based access control (RBAC). Red Hat provides predefined groups and roles to help Organization Administrators manage user permissions.

2.1. USER ACCESS OVERVIEW

You can simplify access management in the Red Hat Hybrid Cloud Console by using role-based access control (RBAC). The User Access feature assigns specific permissions to roles instead of individual users, which are then inherited when roles are assigned to user groups.

You can also create custom groups and roles to provide more fine-tuned control over specific features of Red Hat Lightspeed to suit the needs of your organization.

If you are an Organization Administrator, you can use the User Access feature under **Identity & Access Management → User Access** in the Hybrid Cloud Console to:

- Control user permissions and organize roles.
- Create groups that include roles and their corresponding permissions.
- Assign users to these groups, allowing them to inherit the permissions associated with their group's roles.

All users on your account have access to most of the data in Red Hat Lightspeed.

Additional resources

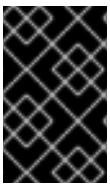
- User Access in the Hybrid Cloud Console
- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP

2.2. PREDEFINED GROUPS IN USER ACCESS

The two predefined groups available in User Access, **Default access** and **Default admin access** provide different baseline privilege levels for users in your organization. You can also create custom groups to align permissions with specific personas, job functions, or teams in your organization.

The Default access group

By default, the **Default access** group is assigned many granular predefined roles, so that group members have basic visibility. Because all users in your organization are members of the **Default access** group, they inherit all permissions assigned to that group. The **Default access** group is automatically updated by Red Hat.



IMPORTANT

If your Organization Administrator modifies the **Default access** group, the group is automatically renamed to **Custom default access**. Once converted, this group is no longer automatically updated by Red Hat.

The Default admin access group

The **Default admin access** group contains only users who have Organization Administrator permissions. This group is automatically maintained, and users and roles in this group cannot be changed.

The **Default admin access** group includes many (but not all) predefined roles that provide update and delete permissions. The roles in this group usually include **administrator** in their names.

TIP

To view the roles included in the **Default access** or **Default admin access** group, log in to the Hybrid Cloud Console, navigate to **Groups** in User Access, and select the group.

Additional resources

- User Access groups

2.3. PREDEFINED ROLES ASSIGNED TO GROUPS

Understand how predefined roles in Red Hat Hybrid Cloud Console bundle permissions across multiple Red Hat Lightspeed applications to align with common user personas. Use predefined roles to reduce administrative effort, or create custom roles for more fine-tuned control over specific features.

The predefined roles are a starting point to help you control and manage user permissions. You can then use these roles to create custom roles that are tailored to your specific use cases and organization. For example, you can use the predefined granular roles to create custom roles that provide more fine-tuned control over specific features of Red Hat Lightspeed.

TIP

Across the Red Hat Lightspeed product documentation, the **Prerequisites** section for each procedure lists which predefined roles provide the permissions needed to use the features in that procedure. For example, if a procedure requires permissions to create and view remediations, the **Prerequisites** section for that procedure lists the *Remediations user* or other valid role as a recommended predefined role to use for that procedure.

Additional resources

- Predefined User Access roles

2.4. CHECK YOUR PERMISSIONS

Verify your current permissions and the roles or groups assigned to you in the Red Hat Hybrid Cloud Console. Check your permissions to troubleshoot access issues or understand your level of access to Red Hat Lightspeed applications.



NOTE

Only users with the Organization Administrator role can view the permissions of other users in the User Access settings and manage user permissions to Red Hat Lightspeed services. For more information, see the *Configure user permissions* section.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ My User Access**.
2. If you try to access Red Hat Lightspeed features and see a message that you do not have permission to perform this action, contact your Organization Administrator or a user with the **User Access administrator** role to request the permissions required to access those features and complete the actions you want to perform.

Verification

All of the applications that you have permissions to access are listed on this page and are grouped by product, for example, RHEL, OpenShift Container Platform, and Ansible Automation Platform.

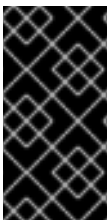
You can also filter your permissions by application, for example, by advisor, cost management, inventory, and remediations.

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

2.5. CONFIGURE USER PERMISSIONS

If you are an Organization Administrator, you can view and manage user permissions for all users in your organization. Control access to Red Hat Lightspeed and other Red Hat Hybrid Cloud Console services through the User Access interface.



IMPORTANT

If you are not an Organization Administrator, you are unable to complete this task. However, you can check your own permissions for different applications by navigating to **Settings → Identity & Access Management→ My User Access**. Contact your Organization Administrator to request more permissions.

Prerequisites

- You have logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator, or you have the required administrator User Access role permissions.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ User Access**.

Verification

From here, you can create and manage:

- **Settings → Identity & Access Management→ User Access → Roles** to determine permissions to Red Hat Lightspeed services and features

- **Settings → Identity & Access Management→ User Access → Groups** to include one or more roles to align with a specific persona, job function, or team in your organization
- **Settings → Identity & Access Management→ User Access → Users** and their assignment to groups to inherit permissions from the roles assigned to those groups

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

2.6. USER ACCESS ROLES FOR PERMISSIONS TO MALWARE DETECTION FEATURES

Understand the predefined roles that control access to malware detection features in Red Hat Lightspeed. Use these role definitions to assign appropriate permissions to users based on their responsibilities.



IMPORTANT

The malware detection detection service does not include a **default-group** role.

To view data or control settings in the malware detection service, users must be members of the User Access group with one of the following roles:

Table 2.1. Permissions provided by the User Access roles

User Access role	Grants permissions to ...
Malware detection administrator	• Read all malware detection data • Set user acknowledgment • Delete hits • Disable signature permissions
Malware detection editor	• Read All • Set user acknowledgment
Malware detection viewer	• Read All

CHAPTER 3. CONFIGURE THE RED HAT LIGHTSPEED MALWARE DETECTION SERVICE

Install and configure the Red Hat Lightspeed malware detection service on RHEL systems, assign User Access roles for malware detection administrators and viewers, run an on-demand scan, and review scan results in the Hybrid Cloud Console.

3.1. MALWARE DETECTION SERVICE SETUP ACTIONS AND REQUIRED ACCESS

To configure and enable the malware detection service, you need specific access and privileges on your Red Hat Enterprise Linux systems.

Some procedures require root privileges on the system and others require that the administrator performing the actions be a member of a User Access group with the **Malware detection administrator** role.

The following table lists the main actions needed to configure the malware detection service, what each step involves, and the required access.

Action	Description	Required privileges
Install YARA	Install the YARA application.	Root privileges
Configure the insights-client	Configure the insights-client to use the malware detection service; enable the collector for the malware detection service.	Root privileges
Configure User Access in the Red Hat Hybrid Cloud Console	In User Access groups, create malware detection groups and then add the appropriate roles and members to the groups.	Organization Administrator role on the Red Hat account
Optional: Enable process scanning	Optionally enable process scanning in **/etc/insights-client/malware-detection-config.yml** by setting **scan_processes** to **true** and running the malware detection collector.	Root privileges
Scan your Red Hat Enterprise Linux systems	Run the malware detection collector scan on your Red Hat Enterprise Linux systems.	Root privileges
View results	See the results of system scans in the Hybrid Cloud Console.	Membership in a User Access group with the **Malware detection viewer** role

3.2. CONFIGURE MALWARE DETECTION ON RHEL SYSTEMS

To start running malware detection scans and reporting data to the Red Hat Lightspeed application, install YARA and the malware detection collector on the RHEL system.

Prerequisites

- The target system runs RHEL 8 or later.
- You have root privileges on the system.

Procedure

1. Install YARA.

   TIP

   Yara RPMs for RHEL 8 and later are available on the Red Hat Customer Portal.

   ```
   $ sudo dnf install yara
   ```

   To confirm that YARA is installed, run the following command from the command line:

   ```
   $ yara --version
   ```

   If YARA is installed, the command displays the version information.

   NOTE

   Red Hat Lightspeed malware detection is not supported on RHEL 7.

2. Register the system with Red Hat Lightspeed.

   IMPORTANT

   Using the malware detection service requires that you have the insights-client package installed on the system and the system registered with Red Hat Lightspeed.

   a. Install the insights-client RPM.

      ```
      $ sudo yum install insights-client
      ```

   b. Test the connection to Red Hat Lightspeed. If the system is properly registered with Red Hat Lightspeed, the command completes successfully and displays a message that the connection test was successful.

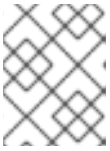
      ```
      $ sudo insights-client --test-connection
      ```

   c. Register the system with Red Hat Lightspeed.

      ```
      $ sudo insights-client --register
      ```

3. Run the insights-client malware detection collector to create a malware detection configuration file in **/etc/insights-client/malware-detection-config.yml**, perform a test scan and upload the results to Red Hat Lightspeed.

```
$ sudo insights-client --collector malware-detection
```



NOTE

This minimal test scan verifies the service is functioning by detecting a few known test matches.

To confirm the test scan completed, navigate to **Security → Malware → Signatures** in the Red Hat Hybrid Cloud Console. You should see a few test matches that are designed to be detected.

4. Perform a full filesystem scan.

   a. Edit **/etc/insights-client/malware-detection-config.yml** and set the **test_scan** option to false.

   ```
   test_scan: false
   ```

   Consider setting the following options to minimize scan time:

   - **filesystem_scan_only** - to only scan certain directories on the system
   - **filesystem_scan_exclude** - to exclude certain directories from being scanned
   - **filesystem_scan_since** - to scan only recently modified files

   b. Run the insights-client --collector again:

   ```
   $ sudo insights-client --collector malware-detection
   ```

   To scan processes after a full filesystem scan, see Enable process scanning for malware detection.

Verification

To confirm that the full filesystem scan completed without errors and the Red Hat Lightspeed malware detection service shows the scan results:

- Navigate to **Security → Malware → Signatures** in the Red Hat Hybrid Cloud Console. The scan results from your full filesystem scan now display.

3.3. ENABLE PROCESS SCANNING FOR MALWARE DETECTION

Extend threat detection by enabling the malware detection collector to scan running processes. When enabled, the collector performs a filesystem scan first, then scans processes for malware.

Prerequisites

- You have root privileges on the system.
- You have a **/etc/insights-client/malware-detection-config.yml** file (created when you first run the malware collector) and have set **test_scan** to **false** which allows a full filesystem scan.

Procedure

1. Edit **/etc/insights-client/malware-detection-config.yml** and set **scan_processes** to **true**.

```
scan_processes: true
```



NOTE

Consider setting these related options while you are changing the configuration for process scanning. These options can help you manage performance when scanning processes, but they are not required to be set to enable process scanning:

- **processes_scan_only** - to only scan certain processes on the system
- **processes_scan_exclude** - to exclude certain processes from being scanned
- **processes_scan_since** - to scan only recently started processes

2. Run the collector again. The collector performs a filesystem scan first, followed by a process scan when **scan_processes** is enabled. When the run completes, view results at **Security → Malware**.

```
$ sudo insights-client --collector malware-detection
```

Verification

- From the command line, locate the **/etc/insights-client/malware-detection-config.yml** file to confirm that the **scan_processes** value is **true**.
- The collector run completes without errors.
- In **Security → Malware** on the Red Hat Hybrid Cloud Console, the latest scan shows the signatures that affect systems and processes.

Additional resources

- View malware detection scan results in the Red Hat Hybrid Cloud Console

3.4. RUN A MALWARE DETECTION SCAN

Run the malware detection collector on a registered RHEL host when you need an on-demand scan. After the scan completes, review the results in the Red Hat Lightspeed malware detection service. Scan time depends on configuration, how much of the system is scanned, and processes included in the scan.

Prerequisites

- YARA and the insights-client are installed and configured on the RHEL system.
- You have sudo access on the system.
- You are logged in to the Red Hat Hybrid Cloud Console.

Procedure

1. To scan a system, run the following command:

```
$ sudo insights-client --collector malware-detection
```

2. View results at **Security → Malware**.

Verification

You can confirm that the scan ran successfully and the results are in the malware detection service by doing the following checks:

- The **sudo insights-client --collector malware-detection** command exits successfully.
- **Security → Malware** shows a new or updated scan for the host.

Additional resources

- View malware detection scan results in the Red Hat Hybrid Cloud Console
- Setting up recurring scans for Red Hat Lightspeed services

3.5. VIEW MALWARE DETECTION SCAN RESULTS IN THE RED HAT HYBRID CLOUD CONSOLE

View results of system scans on the Red Hat Hybrid Cloud Console to see threats that are a risk to your systems.

Prerequisites

- YARA and the insights-client are installed and configured on the RHEL system.
- You have logged in to the Hybrid Cloud Console as a user who is a member of a User Access group with at least the **Malware detection viewer** role.

Procedure

1. Navigate to **Security → Malware → Systems**.
2. View the dashboard to get a quick summary of all of your RHEL systems that have malware detection enabled and are reporting results.
3. To view results for a specific system, use **Filter by name** in the search box.
4. Click the name of a system to view its specific match details.

Verification

You can confirm that you are viewing the correct results for your system by checking the following:

- After you filter by name in **Security → Malware → Systems**, the selected system's detail view matches the system you intended.
- After you click a system name, the malware detection service shows specific match details for that system.

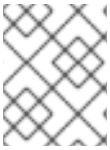
CHAPTER 4. MANAGE SCANNING, SIGNATURES, AND SCHEDULING

Manage malware detection scans, signature rules, and scan scheduling to align with your security policy requirements.

4.1. SYSTEM SCANNING FOR MALWARE DETECTION

The malware detection collector scans the filesystem and optionally runs processes on each RHEL system. You can run scans on-demand from the command line or on a recurring schedule by using **cron**, systemd timers, or automation such as playbooks.

By default, the collector scans all files except virtual filesystems, common mount points, and network filesystem mount points. You can narrow the scope to specific directories or exclude paths by setting options in **/etc/insights-client/malware-detection-config.yml**. Narrowing the scope reduces scan time.



NOTE

Because the scan can take significant time to run, Red Hat recommends running the malware detection scan weekly.

4.2. MALWARE DETECTION SCAN RESULTS

Use the **Signatures** page of the malware detection service to see matched and unmatched IBM X-Force signatures for scans on your Red Hat Enterprise Linux systems.

Matched signatures indicate the YARA software detected a file on a system that corresponds to a known threat. Unmatched signatures indicate the service is actively monitoring for those threats, but no corresponding files were detected during the most recent scan.

The IBM X-Force view on the **Signatures** page lets you filter the signature list to IBM provider content.

4.3. RECURRING MALWARE DETECTION UPLOADS

Configure the malware detection collector to scan on a schedule by using **cron**, systemd timers, or automation. Run **insights-client --collector malware-detection** on a schedule that matches your security policy.

Additional resources

- Setting up recurring scans for Red Hat Lightspeed services

4.4. DISABLE MALWARE SIGNATURES

Disable malware signatures that are not relevant to your organization or cause false positives. This helps reduce noise from intentional configurations, test scans, or matches that do not align with your security priorities.

For example, the signatures **XFTI_EICAR_AV_Test** and **XFTI_WICAR_Javascript_Test** are used to detect the **EICAR Anti Malware** test file and **WICAR Javascript Crypto Miner** test malware. They are intentional test signatures but do not represent actual malware threats. Signatures such as these can be disabled so that matches against them are not reported in the Red Hat Hybrid Cloud Console.

After a signature is disabled, the malware detection service removes any existing matches against that signature from the Hybrid Cloud Console and ignores the signature in future scans. If the signature is re-enabled, the malware detection service again looks for the signature in future malware-detection scans and shows resulting matches.



NOTE

Disabling a signature does not erase the history of previous matches for that signature.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with the **Malware detection administrator** role.

Procedure

1. Navigate to **Security → Malware → Signatures**.
2. Disable signatures by using one of the following methods:
   - **Disable a single signature from the signatures list:**
     a. Find the signature to disable.
     b. Click the options icon (⋮) at the end of the signature row.
     c. Select **Disable signature from malware analysis**
   - **Disable a single signature from the signature details page:**
     a. Find the signature to disable.
     b. Click the signature name.
     c. On the signature details page, click the **Actions** drop-down.
     d. Select **Disable signature from malware analysis**
   - **Disable multiple signatures at the same time:**
     a. Select the checkbox at the start of each row for every signature to disable.
     b. Click the options icon (⋮) next to the filter fields.
     c. Select **Disable signatures from malware analysis**



NOTE

To re-enable a previously disabled signature, follow part of the process to disable the signature, then choose the option to **include the signature in malware analysis** again.

Verification

To confirm that the signature is now disabled, check the following:

- Navigate to **Security → Malware → Signatures**. Confirm each signature you disabled shows as excluded from malware analysis.
- On the same **Signatures** page, confirm matches for that signature no longer appear in active match lists.

Additional resources

- View malware detection scan results in the Red Hat Hybrid Cloud Console

4.5. VIEW DISABLED MALWARE SIGNATURES

If you have access to the malware detection signatures view, you can see how many signatures are disabled and filter the list to show only disabled signatures.

Prerequisites

- You are logged in to the Hybrid Cloud Console as a user who is a member of a User Access group with at least the **Malware detection viewer** role.

Procedure

1. Navigate to **Security → Malware → Signatures**.
2. View the number of disabled malware signatures in the dashboard at the top of the page.
3. Set filters to show the disabled signatures:
   a. Set the primary filter to **Signatures included in malware analysis**
   b. Set the secondary filter to **Disabled signatures**.
4. Optional: Re-enable a previously disabled signature, by using the same pages and menus. Choose the option to include the signature in malware analysis again.

Verification

- Navigate to **Security → Malware → Signatures**. Confirm the dashboard count for disabled signatures matches your expectation after the filter change.
- On the same **Signatures** page, with the primary filter **Signatures included in malware analysis** and the secondary filter **Disabled signatures**, confirm the table lists only signatures that are disabled.

Additional resources

- View malware detection scan results in the Red Hat Hybrid Cloud Console

CHAPTER 5. INTERPRET RESULTS AND MANAGE MATCHES

Understand malware detection results, review signature matches, and manage match status to track your security team's investigation progress.

5.1. INTERPRET MALWARE DETECTION SCAN RESULTS

In most cases, malware detection scans result in no signature matches, meaning YARA found no matching strings or boolean expressions in the scanned files. You can view scan details and match results in the Red Hat Lightspeed malware detection service.

When the malware detection scan with YARA detects a match, it sends the results to Red Hat Lightspeed. You can see match details in the malware detection service UI, including the file and date. System scan and signature match history covers the last 14 days, helping you detect patterns and provide information to your security incident response team. For example, a signature match might appear in one scan but not in the next scan of the same system. That pattern can indicate malware that is detectable only when a certain process is running.

Additional resources

- Malware match status and acknowledgment
- Set the match status of a malware signature

5.2. MALWARE MATCH STATUS AND ACKNOWLEDGMENT

You can acknowledge malware signatures at the system and signature levels, set match status, and reduce noise so your team can focus on real risk.

The **Status** field on the **Signatures** page enables you to select a status for each system or signature that you review. You can change the status of each signature match while you continue investigating and managing malware matches. Status updates help you stay informed about the progress of remediations or evaluations. You can also decide which matches pose low or no threats to your systems or are irrelevant. If you are a member of a group with at least **Malware detection administrator** permissions, you can delete irrelevant matches from your systems.

The **Total Matches** column on the **Signatures** page includes all matches for a signature on a system. You can use the list of matches to track and review the history of malware matches on individual systems in your environment. Red Hat Lightspeed retains malware matches indefinitely unless you delete them. Acknowledging matches and setting their status provides a persistent record of your team's investigation progress. If you delete a system from the malware service, its match records are also deleted.

The **New Matches** column shows the number of new matches for a signature. A bell icon indicates each new match. A new match has a match date of up to 30 days from when the match was detected and has a **Not Reviewed** status. Matches older than 30 days, or those that have already been reviewed, become part of **Total Matches**.

Additional resources

- Disable malware signatures
- Manage user permissions for Red Hat Lightspeed services

5.3. SET THE MATCH STATUS OF A MALWARE SIGNATURE

Assign match status values to malware signature matches as you review findings. Your decisions persist with each match, reducing repeat work when you review the same items in the future.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user with one of the following roles:
  - **Malware detection viewer** - view and filter only
  - **Malware detection editor** - view, filter, and edit
  - **Malware detection administrator** - view, filter, edit, and delete

Procedure

1. Navigate to **Security → Malware → Signatures**. A list of signatures displays at the bottom of the page.
2. Click a signature name. The information page for that signature displays. The page shows the list of systems affected by that malware signature. A bell icon indicates new matches for that signature.
3. Use the filters at the top of the list of affected systems to filter by **Status**. (The default filter is **Name**.)
4. Click the drop-down menu to the right of the **Status** filter and select **Not Reviewed**.
5. Click the drop-down arrow next to the name of an affected system. The list of matches displays, with the most recent matches first.
6. Select the checkbox next to the match that you want to review.
7. To change the status of a match, select the new status from the **Match status** drop-down menu. Select from the following options:
   - **Not reviewed**
   - **In review**
   - **On-hold**
   - **Benign**
   - **Malware detection test**
   - **No action**
   - **Resolved**
8. Optional: Add a note to include more information about the match status. The checkmark indicates that your note registered as saved.

Verification

- The **Match status** value you selected stays on the **Signatures** page for the match after the page updates.
- Optional: If you added a note, the saved state registers as a checkmark.

5.4. DELETE A MALWARE MATCH FROM THE SIGNATURE VIEW

When a match is not relevant to your environment, you can delete it from the list of signatures. This ensures all malware detection administrators and RHEL reviewers see an up-to-date set of open items.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with at least the **Malware detection administrator** role.

Procedure

1. Navigate to **Security → Malware → Signatures**. A list of signatures displays on the lower part of the page.
2. Click the drop-down arrow next to the signature you want to manage. A list of matches displays below the system, with the most recent match first.
3. Click the options icon (⋮) at the far right side of the match you want to delete, and then select **Delete match**. The list of matches refreshes.

Verification

- The deleted match no longer displays in the match list for that signature after the list refreshes.

5.5. VIEW MALWARE MATCHES ON SYSTEMS

View malware matches on individual systems to review which signatures were detected and acknowledge matches so your team's view stays current.

Prerequisites

- You are logged in to the Hybrid Cloud Console as a user who is a member of a User Access group with at least the **Malware detection viewer** role.
- To edit or delete matches, you need to be a member of a group with at least the **Malware detection administrator** role.
- Only systems that have malware detection enabled appear in the list of affected systems. For more information about how to enable malware detection, see Configure the Red Hat Lightspeed malware detection service.

Procedure

1. Navigate to **Security → Malware → Systems**. The list of systems displays. If a system has malware matches, the **Matched** label displays next to the system name.
2. Click a system name. The system details page displays, with the list of matched malware signatures at the bottom.

3. Click the drop-down list next to a malware signature. A list of matches for the signature on the system displays.

4. Acknowledge the matches in the list.

Verification

- After you acknowledge, the UI reflects the acknowledgment for those matches.

Additional resources

- System scanning for malware detection

CHAPTER 6. CONFIGURE THE COLLECTOR AND NOTIFICATIONS

Customize configuration settings for the collector to get the best results from scans of your systems and processes. You can also enable notifications to receive alerts about malware detections on your systems.

6.1. CONFIGURATION OPTIONS FOR THE MALWARE DETECTION COLLECTOR

The **/etc/insights-client/malware-detection-config.yml** file defines scan behavior for the malware detection collector on RHEL systems. The file uses YAML keys for scan scope, exclusions, process scanning, and timing options. You can set the same options by using environment variables when you run the collector. When both variables are present, the environment variables override the file.

Additional resources

- Malware detection collector configuration reference
- Registering RHEL systems and configuring client tools with Red Hat Lightspeed

6.2. MALWARE DETECTION COLLECTOR CONFIGURATION REFERENCE

The malware detection collector reads **/etc/insights-client/malware-detection-config.yml** to determine scan settings, exclusions, and process scanning options. Environment variables set before running the collector override the configuration file.

Table 6.1. Configuration options in malware-detection-config.yml

Option	Description
filesystem_scan_only	Allowlist of files or directories to scan. Only listed items are scanned. Specify one item or a list (YAML list syntax). If empty, all files and directories are scanned (subject to other options).
filesystem_scan_exclude	Denylist of paths not to scan. Default exclusions include virtual filesystems (for example, **/proc**, **/sys**, **/cgroup**), common mount points (**/mnt**, **/media**), and paths such as **/dev** and **/var/log/insights-client**. If a path is in both **filesystem_scan_only** and **filesystem_scan_exclude**, **filesystem_scan_exclude** wins. If a parent directory is in **filesystem_scan_only** but some of its children are in **filesystem_scan_exclude**, the collector scans the parent directory but skips those children.
filesystem_scan_since	Limit file scanning to files modified within a time window. For example, **filesystem_scan_since: 1** scans files created or modified in the past day. Use **filesystem_scan_since: last** for files changed since the last successful **filesystem_scan** of the **malware-client**.

Option	Description
exclude_network_filesystem_mountpoints	When **true** (default), the collector does not scan mount points of network filesystems, reducing traffic and load. Types treated as network filesystems are listed in **network_filesystem_types**. Excluded mount points are merged with **filesystem_scan_exclude**. Set to **false** to allow those mount points unless you exclude them explicitly.
network_filesystem_types	Filesystem types treated as network filesystems for the **exclude_network_filesystem_mountpoints** option.
scan_processes	When **true**, the collector includes running processes in the scan. Default is **false** to limit performance impact on large or busy systems. When**false**, process-scan options are ignored.
processes_scan_only	Allowlist for processes, analogous to **filesystem_scan_only**. Specify a PID, a range (for example, **1000..2000**), or a substring of the process name (for example, **Chrome**).
processes_scan_exclude	Denylist for processes. If a process matches both **processes_scan_only** and **processes_scan_exclude**, **processes_scan_exclude** takes precedence.
processes_scan_since	Time window for process scanning, analogous to **filesystem_scan_since**. For example, **processes_scan_since: 1** limits to processes started or modified in the past day; **processes_scan_since: last** uses the last successful process scan.

Table 6.2. Environment variables for collector options

Variable	Description
General rule	Every YAML option in **/etc/insights-client/malware-detection-config.yml** can be set with an environment variable of the same name in uppercase with underscores. The environment variable overrides the configuration file value when both are present. For example, YAML **test_scan** maps to **TEST_SCAN**.

Variable	Description
FILESYSTEM_SCAN_ONLY, **FILESYSTEM_SCAN_EXCLUDE**, **PROCESSES_SCAN_ONLY**, **PROCESSES_SCAN_EXCLUDE**, **NETWORK_FILESYSTEM_TYPES**	For these list-valued settings, use a comma-separated list in the environment variable (no YAML list syntax). Example: **FILESYSTEM_SCAN_ONLY=/etc,/tmp,/var/lib**.

For an example of passing these variables when you run the collector from the command line, see Configure the malware collector with environment variables.

6.3. CONFIGURE THE MALWARE COLLECTOR BY USING ENVIRONMENT VARIABLES

You can override **/etc/insights-client/malware-detection-config.yml** settings for a single run by exporting or prefixing environment variables when you run the insights-client collector.

Prerequisites

- You have **sudo** access on the RHEL system.
- The insights-client package is installed.

Procedure

1. Set list-valued scan targets with a comma-separated list. For example, to scan only **/etc**, **/tmp**, and **/var/lib**, use the following pattern:

```
FILESYSTEM_SCAN_ONLY=/etc,/tmp,/var/lib
```

2. Run the collector with those variables set and with **TEST_SCAN** disabled, for example:

```
$ sudo FILESYSTEM_SCAN_ONLY=/etc,/tmp,/var/lib TEST_SCAN=false insights-client --collector malware-detection
```

Verification

- The collector command exits with status **0**.
- The scan results or logs reflect the overridden settings for this run. For example, the scan scope is limited to the directories you set in **FILESYSTEM_SCAN_ONLY**.

6.4. NOTIFICATIONS FOR MALWARE EVENTS

Enable the notifications service to receive alerts when the malware service detects signature matches on your systems. Notifications help you stay informed about possible threats when you cannot continually monitor the Red Hat Lightspeed dashboard.

Notifications trigger for Red Hat Lightspeed services based on service-specific criteria.

Malware service notifications include the following information:

- The name of the affected system
- The number of signature matches found during the system scan
- A link to view the details from the Red Hat Hybrid Cloud Console

Enabling the notifications service requires three main steps:

- First, an Organization Administrator creates a role-based access control (RBAC) group that includes at least the **Notifications administrator** role, and then adds account members to the group.
- Next, if you have the notifications administrator role, configure behavior groups for events in the notifications service. Behavior groups specify the delivery method for each notification. For example, a behavior group can specify whether email notifications are sent to all account members or only to Organization Administrators.
- Finally, to receive individual emails for each event, set your notification preferences in the Red Hat Hybrid Cloud Console.

Additional resources

- Configuring notifications on the Red Hat Hybrid Cloud Console with FedRAMP
- Understanding Red Hat Lightspeed Events and Notifications in the Hybrid Cloud Console